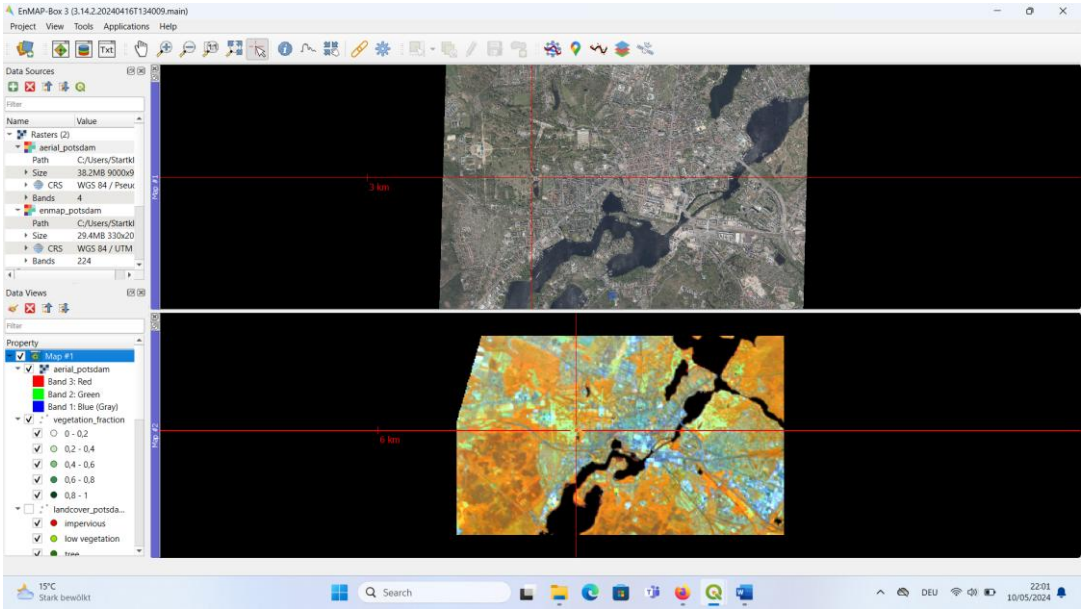
24S856131: Advanced Remote sensing

Assignment: Creating spectral library from hyperspectral and multispectral imagery

Task 1

1.1. The two images are displayed in the map wiever window as follows. “Aaerial_postdam” Image has only 4 bands while the “enmap_postdam” image has 224 bands. Multispectral images have lower number of bands while hyper spectral images have higher number of bands. Therfore the enmap-postdam imgae is the hyperspectral image and the aerial-postdam image is the multispectral image.

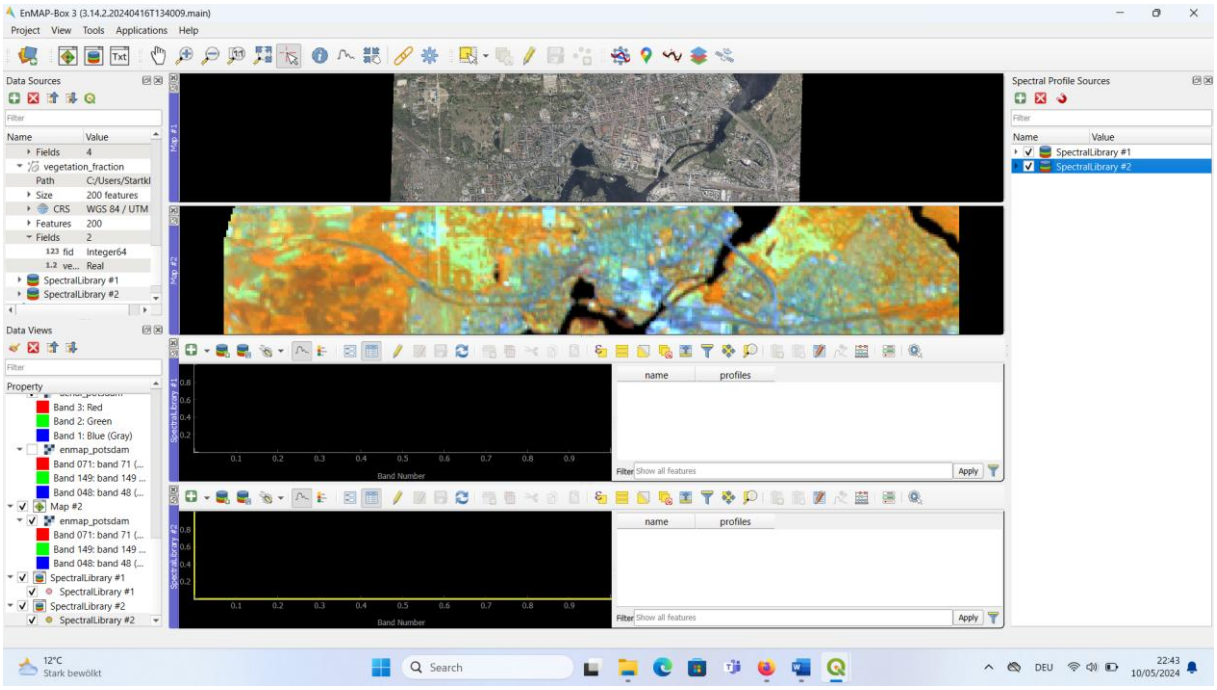


1.2 and 1.3). The spectral range of the hyperspectral image - i.e. enmap-postdam is between 418.24 nm – 2445.53 nm while the spectral resolution is 330 x 200 pixels.

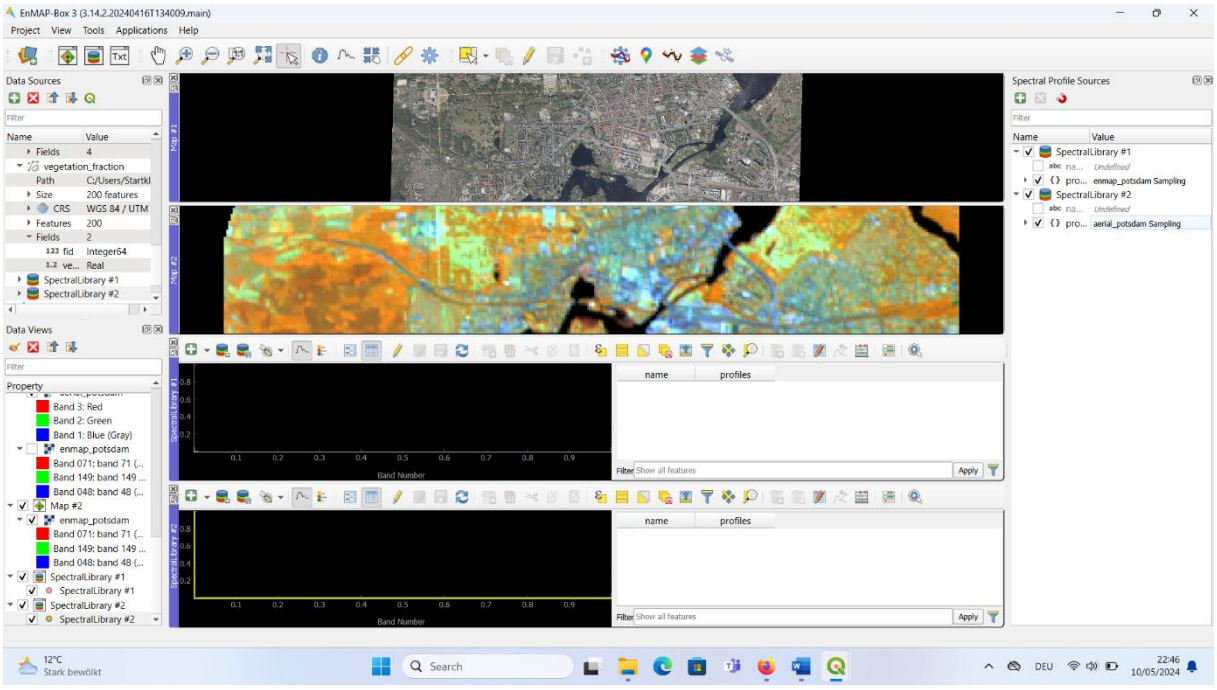
Task 2

<u>Create a Spectral library using hyperspectral and multispectral images.</u>

Step 1

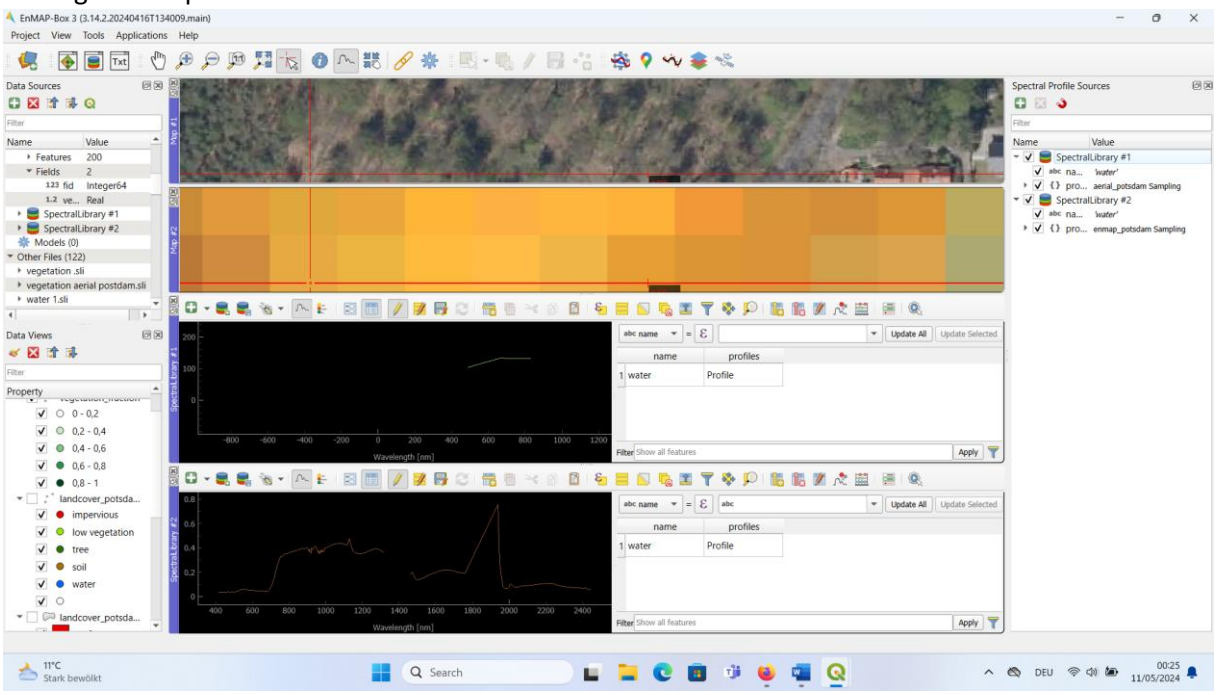


Step 2



Step 3

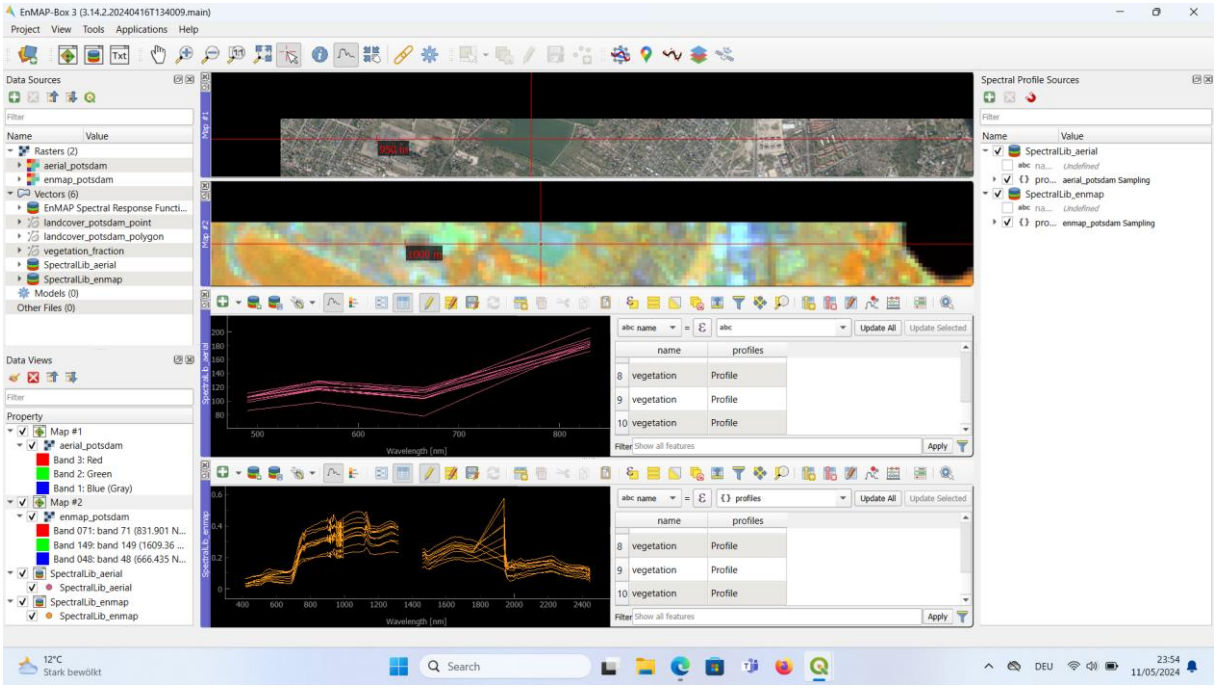
Linking two maps



Spectral libraries from the multispectral and hyperspectral image

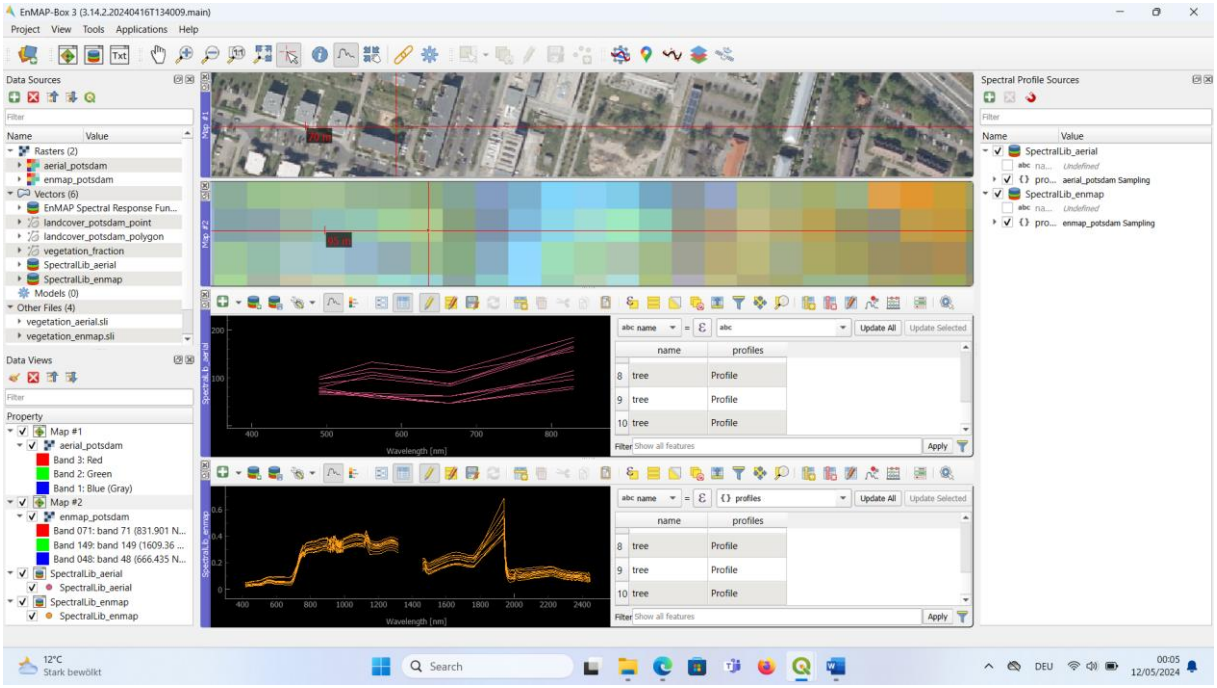
Spectral library for the aerial image(multispectral image) only shows spectral profiles within the wave length range of 490 nm - 830 nm while the enmap image (hyperspectral image) shows a wider wave length range between 424 nm - 2445 nm. In every spectral library created from the hyperspectral image, there is a spectral profile gap between the wave length range 1350 nm-1450 nm. And also there is a peak of reflectance in between the wave length range of 1900-2000 nm for all the features considered here. Furthermore the reflectance pattern between the wave langth range of 900-1000 nm is quite similar for all the features in the hyperspectral image.

Spectral libraries for Vegetation



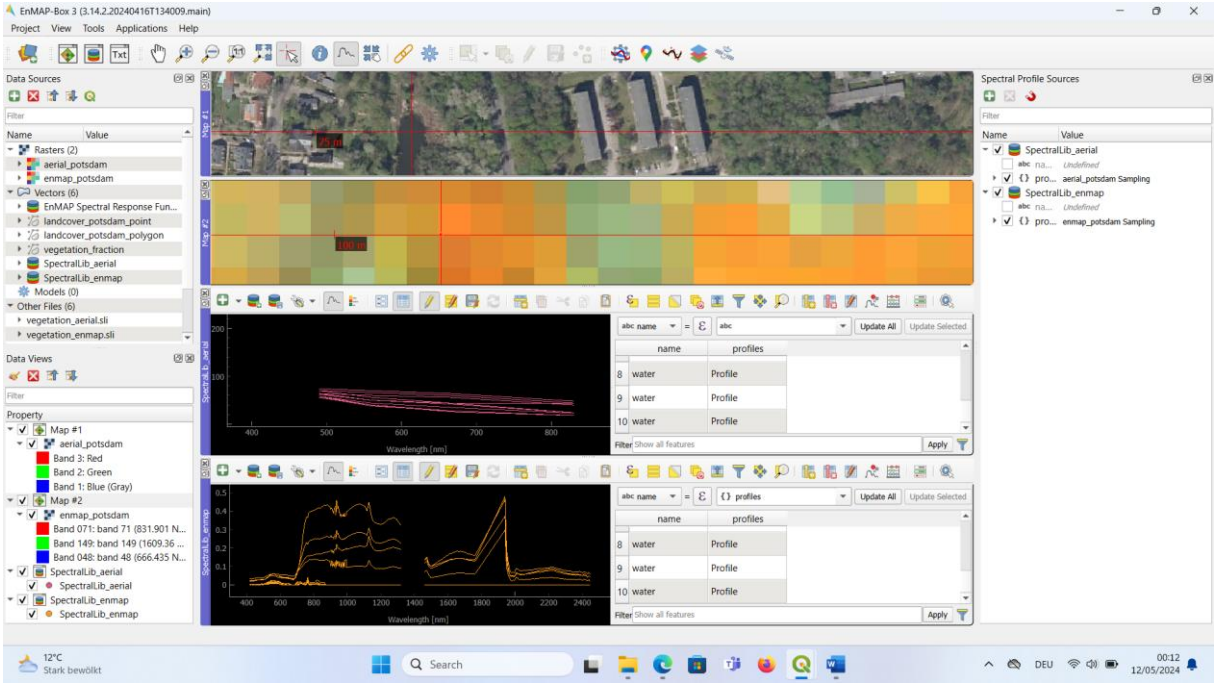
Spectral profiles of both the images show similar reflectance pattern upto the wave length range they both represent together. There is more reflectane close to the wave length 570 nm and more absorption close to 680 nm in both spectral profiles. However the multispectral image shows a more linear pattern while the hyperspectral image shows a curvy shape in the spectral profiles. Almost all the spectral profiles for similar class features are more or less similar in their spectral profile shapes too.

Spectral libraries for trees



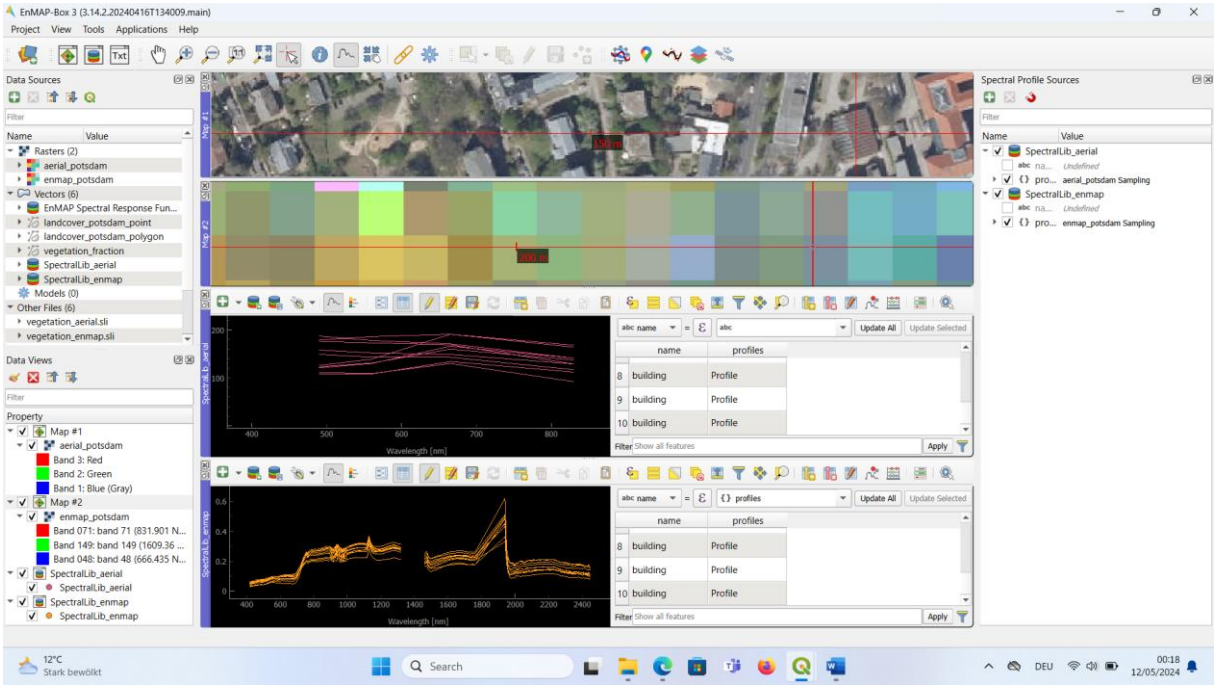
It is observed that the spectral profiles of trees are also very similar to each other and to the spectral profile of vegetation for both image groups.

Spectral libraries for water

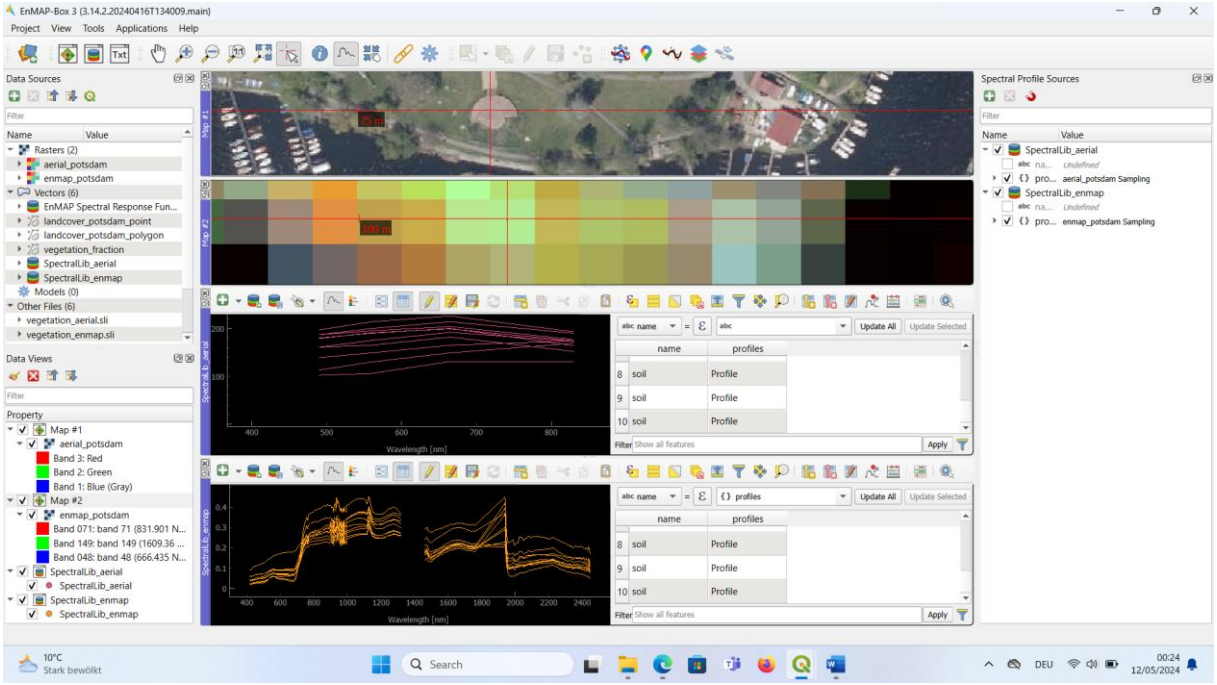


Here water shows a very low reflectance according to the spectral profile of the multi spectral image. However, it does not represent by the hyperspectral image.

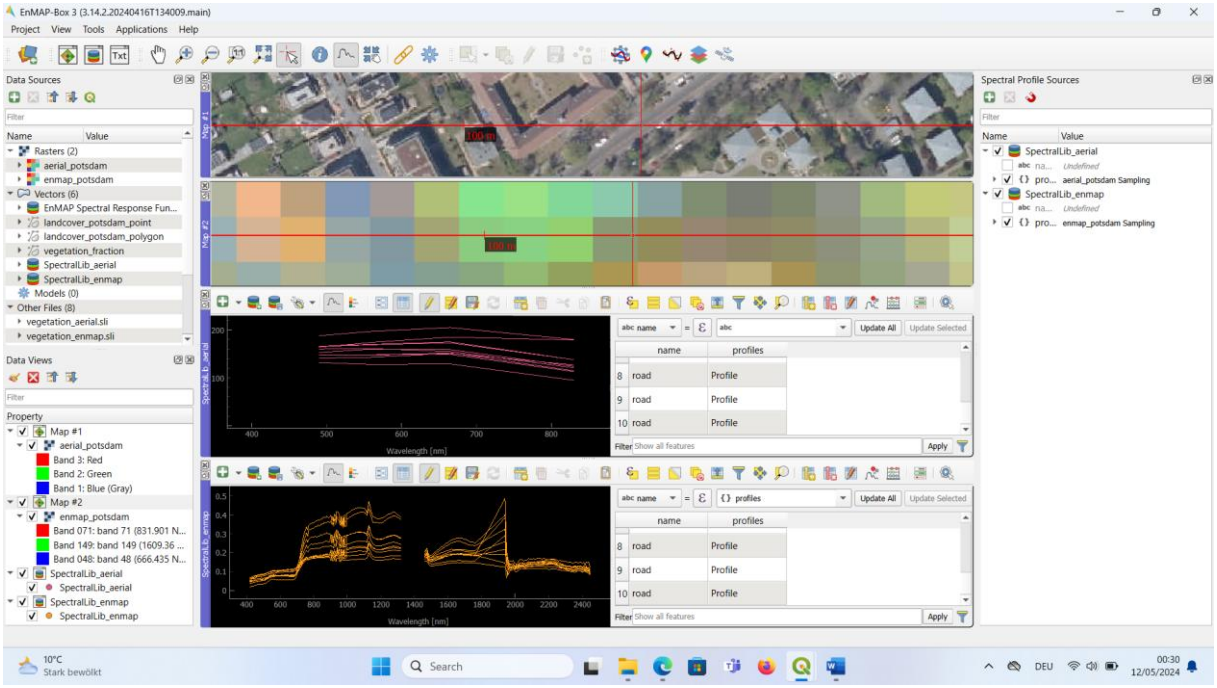
Spectral libraries for buildings



Spectral libraries for soil

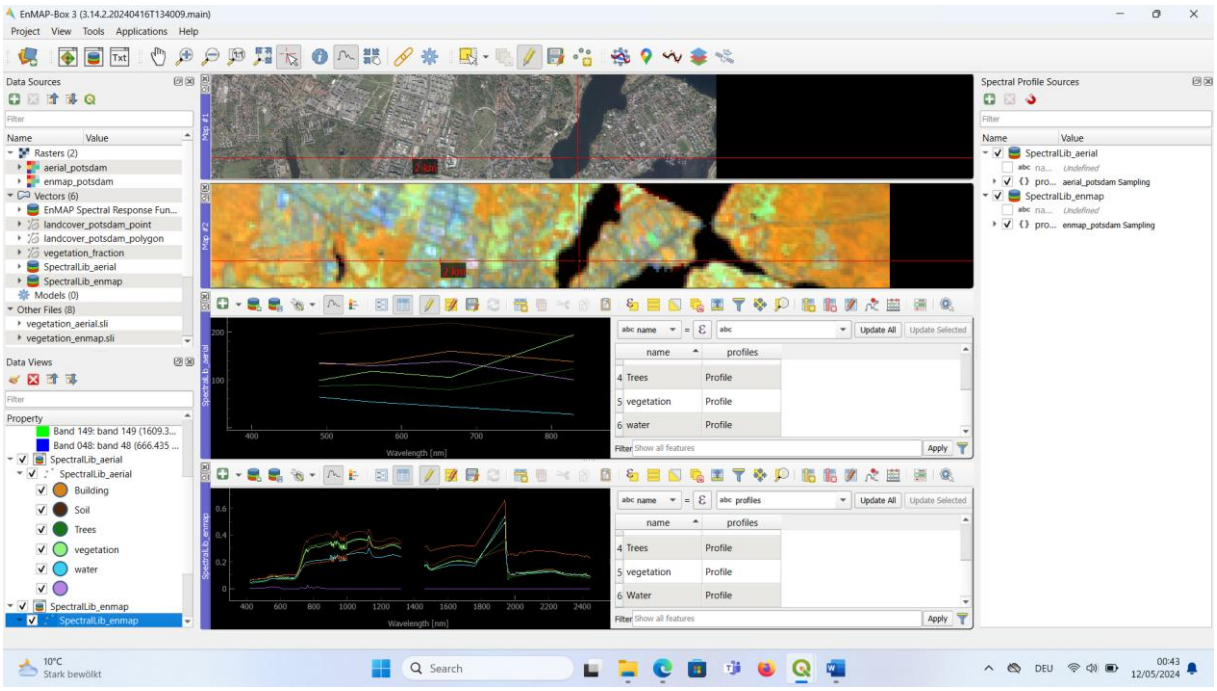


Spectral libraries for roads



Spectral profiles of soil and roads go together in their pattern and in the reflectance level in both the images.

Spectral libraries for all features



It can be concluded that the hyperspectral images give more detailed and unique spectral profiles due to their broad wavelength range and the higher number of bands than those of the multispectral images. And multispectral images do not provide that uniqueness in their spectral profile presentations.